Fiscal Year:	FY 2011	**Task Last Updated:**	FY 10/27/2011
PI Name:	Bingham, Gail E Ph.D.		
Project Title:	Validating Vegetable Production Unit Plants, Protocols, Procedures and Requirements Using Currently Existing Flight Resources		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Advanced human support technologies		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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City:	North Logan	**State:**	UT
Zip Code:	84341	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-01
Start Date:	01/01/2004	**End Date:**	09/29/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	10
No. of Bachelor's Candidates:	6	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: sometime in 2009, new grant was awarded with end date of 9/29/2011, per PI ; NNX09AV53G added. (Ed., 10/27/2011) NOTE: per PI, project transferred to KSC management (5/2008 report) NOTE: Per PI, end date is 7/31/2011; changed 8/24/2007 NOTE: corrected end date to 07/31/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:	Addition of Shane Topham as Co-PI/program manager.		
COI Name (Institution):	Garland, Jay (Life Support Group, Kennedy Space Center) Bates, Scott (Department of Psychology, Utah State University) Bugbee, Bruce (Department of Plants, Soils and Biometeorology, Utah State University)		
Grant/Contract No.:	NNX09AV53G ; NNJ04HG03G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	NASA and the other space agencies are considering including a Vegetable Production Unit (VPU) as a component of their space flight vehicles, including the Crew Exploration Vehicle (CEV) that is planned to carry humans from the Earth to Mars. As the United States moves to define the CEV, critical questions about the role of a VPU and its optimum design and exploitation remain unanswered. Two critical questions revolve around the quantification of the non-nutritional benefits of plants in the cabin and the determination of the operational conditions that maximize these benefits. Plants not only provide food, but for many they provide comfort and relaxation – a diversion from the stress of required activities. These values are currently based only on anecdotal and untested observations that need verification. Part of the benefit may be a small fresh food source, which makes food safety issues important. The VPU Plant, Procedure, Protocol, and Food Safety Requirement (P3R) study, funded under NASA award number NNJ04HG03G, addresses these critical path risks, and, while monitoring microbial accumulations around the on-orbit plant growth facility (Lada), also contributes to the knowledge of crew health risks and their reduction. The Lada VPU P3R project was awarded under the NASA Research Announcement 03-OBPR-01-0000-0111. The project was assigned to the Applied Technology Flight Program Research Integration Office (KT-B) at Kennedy Space Center (KSC) for sample return development and integration. 2. LADA VPU P3R STUDY OBJECTIVES The objectives of this study were to investigate the following: Objective – Non-nutritional Value: Quantify the mass value that should be assigned to the non-nutritional effects of plants in a spacecraft cabin. How much vegetable yield is required for a measurable boost to crew response under the isolation and stress of long-term space missions? How should this stress-relieving resource be presented (open or closed, single crew tended or multiple access) to maximize its value? Is this resource of value to all crew members, or is there a subset that prizes the experience much more highly? Objective – Food Safety: Identify the threat levels and validate the procedures and protocols required to allow US astronauts to eat space-grown vegetables. Determine how to implement these procedures to maximize crew mental health benefits with minimum mission costs. What procedures and hardware need to be in place to assure crew health on a long flight? Objective – Plants: Identify, flight-qualify, and optimize the support requirements for the varieties of vegetables and flowers that are most likely to meet flight crew needs (wants) and that fit within the hardware resource limitations that exist on the ISS and will exist on the CEV. This work will validate the technology readiness level of our available cultural practices to provide reliable, low-cost, stimulating products for crew wellbeing. What plants can tolerate cabin ethylene levels without adding significant cost? How well do seeds need to be protected to survive and flourish on the interplanetary trip? Will seed-producing plants be stable over the 500- to 600-day mission? The objectives were structured to provide the data necessary to both determine whether VPU hardware should be considered for the CEV and to qualify plant products for authorized consumption on the ISS and the CEV by US crewmembers. The Lada hardware and our cooperative agreement with IBMP have provided an ideal, continuing opportunity to investigate many of the plant, procedure, protocol, and food safety requirement (P3R) issues associated with meeting crew livability requirements.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	SPACE APPLICATIONS Plants not only provide food, but for many they provide comfort and relaxation – a diversion from the stress of required activities. In the past, NASA has researched the option of including a Vegetable Production Unit (VPU) as a component of the life support system of their long-duration space flight vehicles. A previous road mapping exercise identified priority 1 and 2 Critical Path Risks for which plant materials and growth support activities could provide solutions. For at least some people plants provide significant non-nutritional benefit to long-duration crews. These values are currently based only on anecdotal and untested observations that need verification. However, we recently launched a ground astronaut survey to better verify these theories. Part of the benefit may be a small fresh food source, which makes food safety issues important. The VPU study addresses these Critical Path risks, and in the process of monitoring microbial accumulations around Lada also contributes to crew health risk knowledge and reduction. EARTH APPLICATIONS Understanding the oxygen to water balance in the root zone applies not just in space, but in terrestrial agriculture as well. There is an acute need to optimize the efficiency of water use in agriculture, and leach-free experiments can be conducted only in space. This project will facilitate the understanding of how to better irrigate (precision watering) and fertilize different plants. Effects of LED lighting were investigated, and the research resulted in the selection of an ideal spectrum on plant growth and development. The experiment also has considerable environmental implications as the ISS cabin is the ultimate setting for practicing recycling of plant, soil, water, and atmospheric resources.
	OVERVIEW OF 2011 RESEARCH OPERATIONS (October 2011 Reporting) With the conclusion of International Space Station experiments the team focused their efforts in the final year of the program on furthering the program objectives with a range of ground work as follows. Food Safety work reached its culmination this year through the efforts of KSC researchers in the completion of a HACCP plan encompassing foods grown in space for human consumption. The HACCP plan is included as an appendix to this report. Non-nutritional value work focused on trying to obtain and analyze data from the crew surveys and the Russian Mars500 experiment. Despite continued attempts to obtain crew survey data, no voluntary crew surveys were ever provided by former crews. Through our collaboration efforts with the Institute for Biomedical Problems in Moscow Russia, significant psychology data from that experiment was made available to USU researchers, and that data was analyzed and included in an original paper. Plants work focused on optimizing support requirements by thoroughly investigating new white LED technologies for

Task Progress:	plant lighting. A very detailed and broad ground investigation resulted in very interesting data on a variety of crops grown under a variety of LED spectra that through phosphors simulate existing compact fluorescent lighting being used. Effects of LED lighting were investigated and the research resulted in the selection of an ideal spectrum for use as a future lighting source in plant growth hardware for space applications. The studies show that critical resources in space plant growth such as power, and volume can be significantly optimized by this new technology. Results also showed that spectrum selection can be used to optimize plant growth and utilization of small contained environments as well as to maximize food production. The accomplishments of the VPU research team during the 2011 year are as follows: • Completion of HACCP plan • Completion of analysis of MARS 500 plant-human interaction psychology data • Completion of ground study of the effect of white LED light sources on plant growth and development REPORTED IN NOVEMBER 2010: The accomplishments of the VPU research team during the 2010 year are as follows: • Two plant growth experiments completed on ISS during expeditions 21 and 22 with crops of wheat and mizuna grown simultaneously in the same root module during expedition 21 and two crops of wheat grown simultaneously in duplicate growth chambers during expedition 22. • Frozen samples from three crops of space grown mizuna returned by shuttle from ISS and analyzed • Two spent root modules returned by shuttle from ISS after taking being used for two chamber experiment, and analyzed for microbial growth, nutrient release, and root media conditions, A21 and A22 • New recommendations for root module packing and fertilizer tested for verification in space and on ground • Successful collection of psychological data during start of Mars habitation chamber 500-day study in Moscow • Approvals obtained for on-orbit crew survey and post-flight crew survey of spaceflight participants who have worked with any plants in space • Successful ground control experiment completed at KSC to grow mizuna under replicated ISS environmental conditions during two chamber experiment on ISS • Construction began on two new Lada root modules A23 and A24 The Lada VPU P3R experiment proposed schedule for downmass and frozen sample support as follows: • One root module and one frozen mizuna sample returned Nov 27, 2009, on STS-129 • Start of the first on-station, side-by-side, two-chamber plant growth experiment to evaluate root module packing improvements began on Feb 13, 2010 • Two root modules and two frozen plant tissue samples, one refrigerated plant tissue sample returned Apr 20, 2010, on STS-131 • Participation in Mars 502 study with plants in the simulation chamber starting on June 3, 2010
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Abstracts for Journals and Proceedings	Bates SC, Gushin VI, Vinokhodova AG, Sychev VN, Bingham GE. "Assessing Potential Non-Nutritive Benefits of Growing and Tending Plants During the MARS 500 series of Chamber Simulations." Paper presented at the 26th annual meeting of the American Society for Gravitational and Space Biology, National Harbor, MD, November 4-7, 2010. Program and Abstracts, 26th Annual Meeting of the American Society for Gravitational and Space Biology, National Harbor, MD, 2010. , Nov-2010
Abstracts for Journals and Proceedings	Bates SC, Gushin VI, Vinokhodova AG, Bogoev B, Sychev VN, Bingham GE. "Countermeasures: Non-nutritive benefits of growing and tending plants during the 500-day chamber simulation." 61st International Astronautical Congress, Prague, Czech Republic, September 2010. 61st International Astronautical Congress, Prague, Czech Republic, September 2010. , Sep-2010
Abstracts for Journals and Proceedings	Schlact IL, Ono A, Bates SC, Peldszus R, Masali M, Rötting M, Foing B, Westernberg A, Stoker C, Stricker FL. "Mars habitability project at MDRS: Sensory experience and creative performance for manned planetary exploration." 61st International Astronautical Congress, Prague, Czech Republic, September 2010. 61st International Astronautical Congress, Prague, Czech Republic, September 2010. , Sep-2010
Abstracts for Journals and Proceedings	Schlacht IL, Ono A, Masali M, Bates SC, Buzzoni L. "Quality of life in space: Life on Earth as simulation to the psychological stability in space." Paper presented at the Workshop on Human Behaviour and Performance in Analogue Environments and Simulations, Noordwijk, Netherlands, December 7-8, 2009. Human Behaviour and Performance in Analogue Environments and Simulations. December 2009. , Dec-2009
Abstracts for Journals and Proceedings	Bates SC, Gushin VI, Vinokhodova AG, Sychev VN, Bingham GE. "Plants as countermeasures: Non-nutritive benefits of growing and tending plants during the initial, 105-day, chamber simulation of MARS 500 experiment." Presented at the 18th IAA Humans in Space Symposium, Houston, Texas, April 11-15, 2011. 18th IAA Humans in Space Symposium, Houston, Texas, April 11-15, 2011. , Apr-2011
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Articles in Peer-reviewed Journals	Cope KR, Bugbee B. "Spectral effects of light-emitting diodes on plant growth and development: Absolute versus relative amounts of blue light." Submitted for publication to HortScience, May 2011. , May-2011
Articles in Peer-reviewed Journals	Cope KR, Bugbee BG. "Spectral effects of light-emitting diodes on plant growth and development: The importance of green light." Submitted for publication to HortScience, June 2011. , Jun-2011
Articles in Peer-reviewed Journals	Shagimardanova E, Gusev O, Bingham GE, Levinskikh MA, Sychev VN, Tiansu Z, Kihara M, Ito K, Sugimoto M. "Oxidative stress and antioxidant capacity in barley grown under space environment." Biosci Biotechnol Biochem. 2010 Jul 23;74(7):1479-82. PMID: 20622437 , Jul-2010
Articles in Peer-reviewed Journals	Bates SC, Marquit J. "Space psychology: natural elements in habitation design." Personal and Ubiquitous Computing. 2011 Jun;15(5):519-23. http://dx.doi.org/10.1007/s00779-010-0316-6 , Jun-2011
Books/Book Chapters	Bates SC, Marquit JM. "Psychology: Natural elements as well-being stimuli in outer space." in "Lunar Settlements." Ed. H. Benaroya. Boca Raton, Fla. : CRC Press, 2010. p. 215 - 224. (Advances in Engineering), Sep-2010
Papers from Meeting Proceedings	Hummerick ME, Garland JL, Bingham G, Sychev VN, Podolsky IG. "Microbiological analysis of Lada Vegetable Production Units (VPU) to define critical control points and procedures to ensure the safety of space grown vegetables." Presented at the 40th International Conference on Environmental Systems, Barcelona, Spain, July 11-15, 2010. 40th International Conference on Environmental Systems, Barcelona, Spain, July 11-15, 2010. AIAA paper AIAA-2010-6255. , Jul-2010